

Projects: Double descent and averaged SGD inference

00103335: Deep Learning and Reinforcement Learning — Fall 2026

Method (a), (c). For each $s \in \{5, 2, 1\}$, set $\beta = \sqrt{s} e_1$, $n = 200$, and use

$$p \in \{20, 40, 60, 80, 100, 120, 140, 160, 180, 199, 200, 220, 240, 260, 300, 400, 600\}.$$

For every (s, p) , perform 20 independent repetitions. In each repetition, draw fresh, mutually independent $X \in \mathbb{R}^{200 \times p}$, $\varepsilon \in \mathbb{R}^{200}$, and $X_{\text{test}} \in \mathbb{R}^{2000 \times p}$ with independent standard Gaussian entries. Form $y = X\beta + \varepsilon$, compute the minimum-norm solution $\hat{\beta} = X^+ y$ using `numpy.linalg.lstsq`, and record

$$R = \frac{1}{2000} \|X_{\text{test}}(\hat{\beta} - \beta)\|_2^2.$$

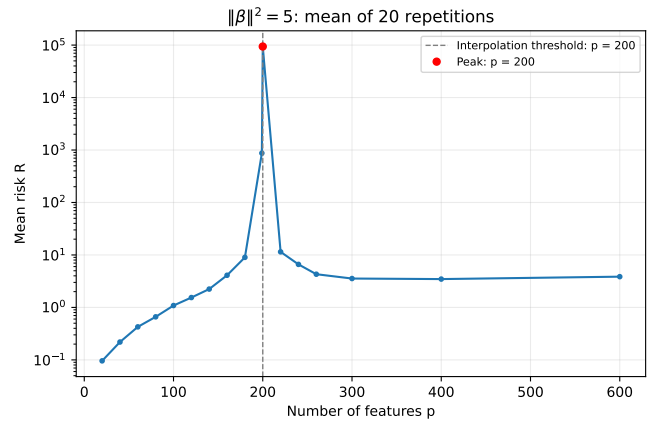
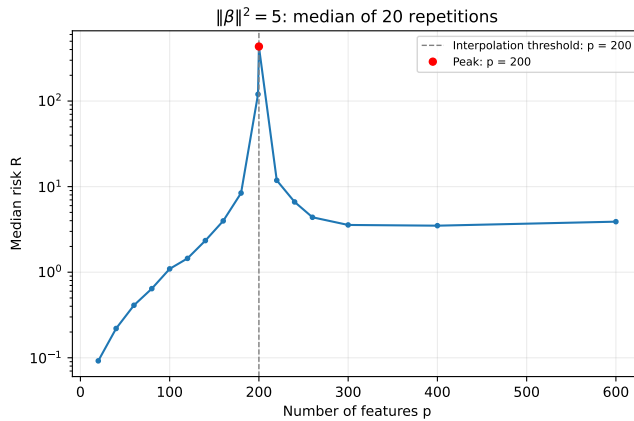
The test target is noiseless, as specified. The random seed is 20260914; `results/risks.csv` contains all 1,020 risks.

Peaks (b), (c). Each plot below uses the 20 recorded risks at each p . The vertical axis is logarithmic; medians and arithmetic sample means are computed on the original risk scale. Their observed maxima are:

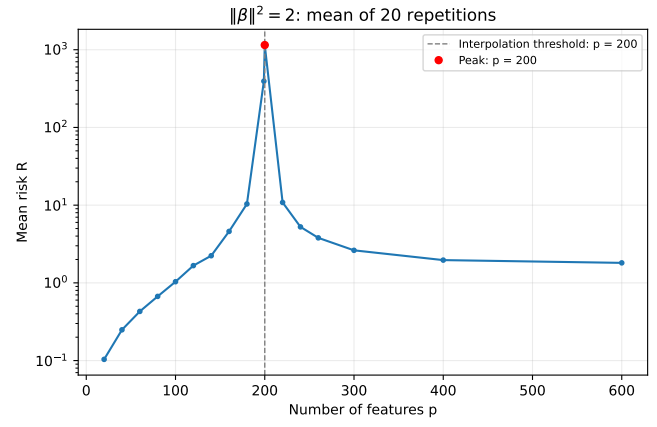
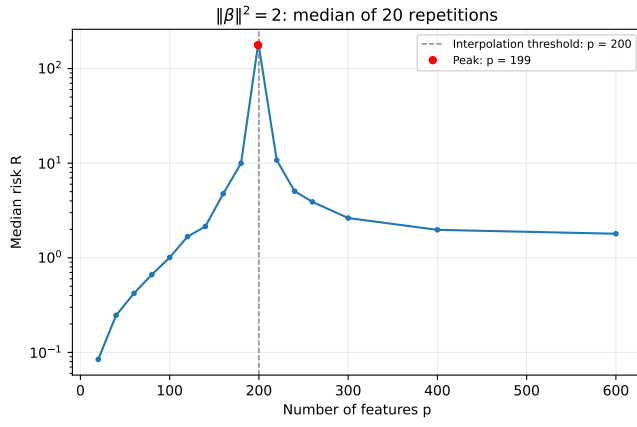
β	Median peak p	Median risk	Mean peak p	Mean risk
$\sqrt{5} e_1$	200	433.826	200	93888.507
$\sqrt{2} e_1$	199	177.463	200	1149.604
e_1	200	448.124	200	5586.063

Risk rises sharply near the interpolation threshold $p = n = 200$ and falls on the overparameterized side. For $\beta = \sqrt{2} e_1$, the observed median peaks at 199, slightly above the median 170.975 at 200. These are finite-sample peaks from 20 repetitions. The much larger mean than median near the threshold reflects extreme risks in some repetitions.

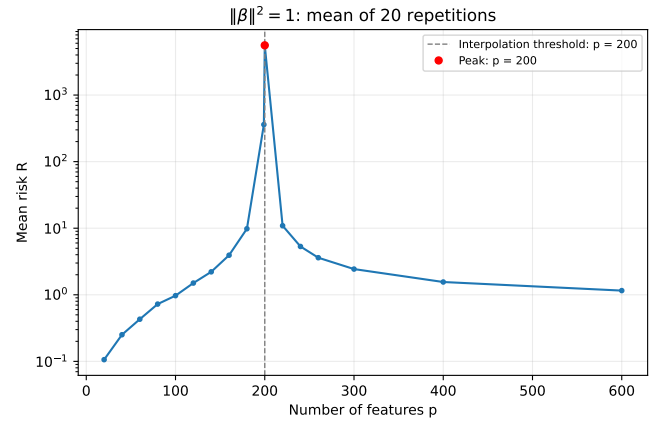
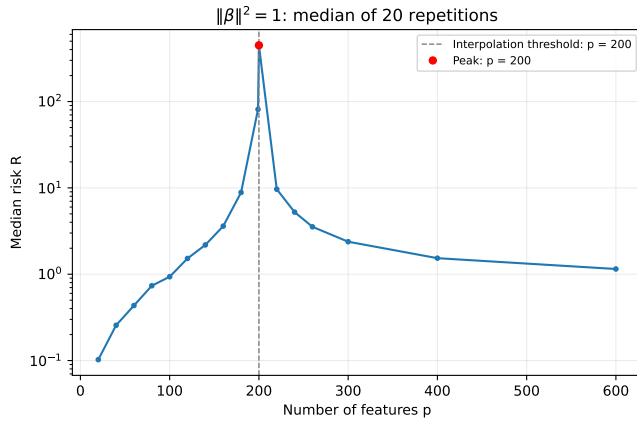
(b) $\beta = \sqrt{5} e_1$.



(c) $\beta = \sqrt{2}e_1$.



(c) $\beta = e_1$.



Problem 2: Asymptotically valid inference for averaged SGD

(a) **Gradient, Hessian, and minimizer.** Set $u = x^\top \theta$. Differentiating the logistic loss gives

$$\nabla_\theta \ell = (\sigma(u) - y)x, \quad \nabla_\theta^2 \ell = \sigma(u)(1 - \sigma(u))xx^\top.$$

Indeed, $d \log(1 + e^u)/du = \sigma(u)$ and $\sigma'(u) = \sigma(u)(1 - \sigma(u))$. Under the stipulated Bernoulli model,

$$\nabla L(\theta^*) = \mathbb{E}[(\sigma(X^\top \theta^*) - Y)X] = \mathbb{E}[\{\sigma(X^\top \theta^*) - \mathbb{E}(Y | X)\}X] = 0.$$

The Gaussian covariate has positive-definite covariance Σ_X (eigenvalues 0.6 and 1.4). For any nonzero v ,

$$v^\top \nabla^2 L(\theta)v = \mathbb{E}[\sigma(X^\top \theta)(1 - \sigma(X^\top \theta))(v^\top X)^2] > 0,$$

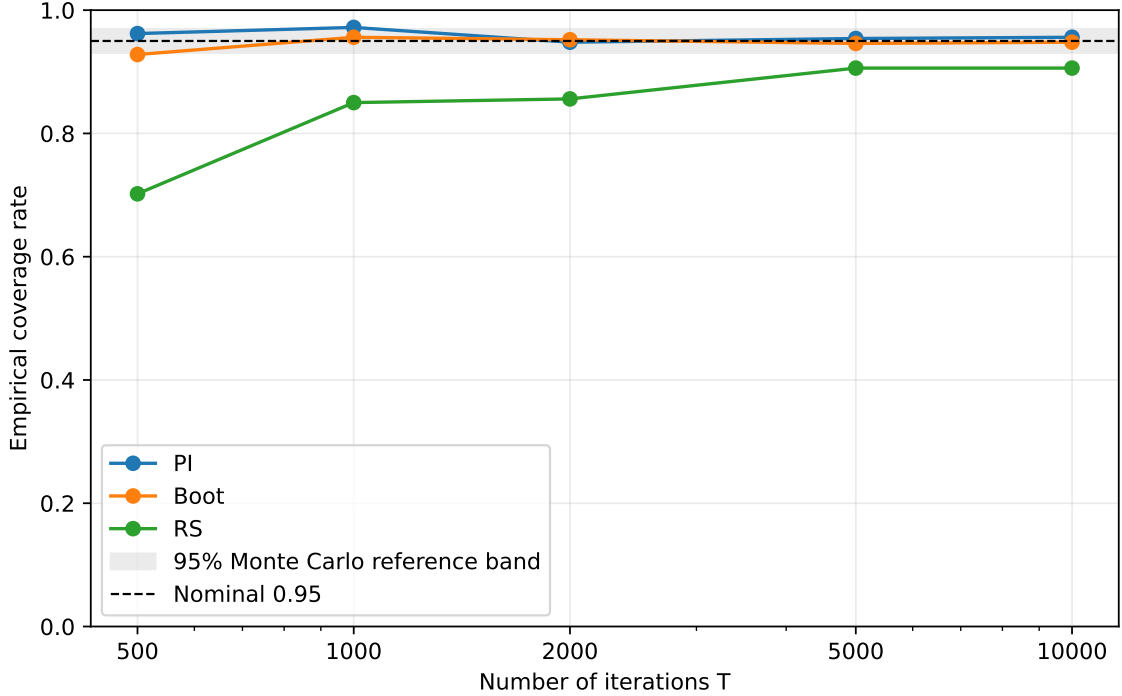
since the logistic factor is positive and $v^\top X$ is a nondegenerate Gaussian. Gaussian moments justify differentiation under the expectation. Thus L is strictly convex, and its stationary point θ^* is its unique minimizer. The contrast is $\psi^* = c^\top \theta^* = -1/\sqrt{5}$.

(b) **Online computation.** Each of 500 independent outer replications generates one fresh $X_t \sim N(0, \Sigma_X)$ and one conditional Bernoulli Y_t at each $t = 1, \dots, 10000$. The observed SGD path starts at zero and uses $\eta_t = 1.6(t + 10)^{-0.6}$. Within a replication, all 200 bootstrap paths use this same data stream and independent $\text{Exp}(1)$ multipliers. We retain θ_t only for $t > 200$ and evaluate all three stated intervals at the five specified checkpoints. For the plug-in interval, we accumulate the full Hessian and outer products of the gradients computed at θ_{t-1} . We evaluate $c^\top \hat{H}_T^{-1} \hat{S}_T \hat{H}_T^{-1} c$ without replacing the sandwich by $\hat{H}_T^{-1}$. For random scaling, if $P_s = \sum_{j=1}^s c^\top \vartheta_j$, we compute

$$\hat{r}_T = \frac{1}{n_T^2} \sum_{s=1}^{n_T} (P_s - s\hat{\psi}_T)^2,$$

which is algebraically the stated formula. All calculations use double precision. Outer replication r uses seed $20260923 + r - 1$; results and individual intervals are saved in the accompanying CSV files.

(c) **Empirical coverage.** The figure plots coverage of the true contrast over 500 independent replications. The shaded band is $0.95 \pm 1.96\sqrt{0.95(0.05)/500}$.



T	Plug-in	Bootstrap	Random scaling
500	0.962	0.928	0.702
1000	0.972	0.956	0.850
2000	0.948	0.952	0.856
5000	0.954	0.946	0.906
10000	0.956	0.948	0.906

At $T = 10000$, the mean interval lengths are 0.08407 (plug-in), 0.08424 (bootstrap), and 0.09871 (random scaling). The plug-in and bootstrap coverages at this checkpoint lie within the Monte Carlo reference band $[0.931, 0.969]$; random scaling coverage is below it. These are the observed finite-sample results with the stipulated critical values and step sizes; no tuning or seed selection was applied.